INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SLEEVING, TEXTILE, BRAIDED, SYNTHETIC POLYMER, -55 TO +105EC

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers one type of black, tubular, braided sleeving, in fourteen (14) sizes, which is used as protective sheathing over internal cable and wire bundles and harnesses in electronic equipment. This sleeving is made with a synthetic polymer, such as polyamide, and is suitable for use within a temperature range of -55EC to +105EC.

2. SALIENT CHARACTERISTICS.

2.1 Material. The sleeving shall be braided using cordage made from a synthetic polymer, preferably polyamide, which will permit the finished product to conform to the salient characteristics of this CID. Cordage shall be made by twisting together two or more ply yarns; ply yarns shall be made by twisting together two or more yarns; and yarns shall be made by twisting together two or more single continuous filament threads of the synthetic polymer. Joints/splices in the sleeving shall be limited to fused joints in yarn or cordage only. Joints shall not appear more often than once in 25 yards of sleeving, and not more often than 20 times in 600 yards.

2.2 Sleeving dimensions. Sleeving shall conform to the dimensions and construction details identified in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to: Commander, Defense Supply Center Richmond, DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

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TABLE I. Sleeving dimensions and construction characteristics.

Sleeving size	Inside diameter (inches, nominal)	Number of ends	Number of carriers	Picks per inch	Wall thickness (inch)
1/8	1/8	24	24	20 ± 2	0.030 + 0.015
3/16	3/16	32	32	18 ± 2	
1/4	1/4	44	44	20 ± 2	
5/16	5/16	44	44	18 ± 2	
3/8	3/8	48	48	18 ± 2	
1/2	1/2	64	64	16 ± 2	
5/8	5/8	128	64	12 ± 2	0.040 + 0.015
3/4	3/4	128	64	11 ± 2	
7/8	7/8	192	64	10 ± 2	
1	1	216	72	10 ± 1	0.050 + 0.015
1-1/4	1-1/4	216	72	9 ± 1	
1-1/2	1-1/2	288	72	8 ± 1	0.060 + 0.015
1-3/4	1-3/4	288	72	7 ± 1	
2	2	360	72	6 ± 1	

2.3 Color. Unless otherwise specified, the sleeving shall be colored gloss black using a chrome fast dye.

2.4 Braid coverage. The braid coverage shall not be less than 85 percent (see 4.6.5).

2.5 Tensile strength. The minimum breaking strength of jointless cords in the sleeving shall be not less than 25 pounds per square inch. The minimum breaking strength of fused cord joints shall not be less than 70 percent of the average breaking strength of the jointless cords.

2.6 Fungus resistance. The sleeving shall not support the growth of fungus.

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2.7 Temperature. The sleeving shall be suitable for use as a protective sheath for internal electrical cable and wiring bundles and harnesses in spacecraft, aircraft, ship, ground vehicle, and shore-based electronic equipment throughout the temperature range of -55EC through +105EC.

2.8 Workmanship. The sleeving shall be free of visible defects, dirt, grime, grease, and abraded or damaged surfaces.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer=s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The contractor must be able to show a history of successful sales of the same or similar items to the commercial market or to the Government for a period of at least two years.

4.3 Sleeving inspection. The sleeving shall be inspected (examined and tested) in accordance with paragraphs 4.4 through 4.6.8 below, unless otherwise excluded, amended, or modified by the contract or order. Unless otherwise specified in the contract or order, these inspections shall be performed on each production lot. When inspection reports are required by the contract or order, they shall contain the individual values measured and the results of all non-measurement examinations.

4.4 Test conditions. Unless otherwise specified herein, inspection of sleeving shall be made under the following environmental conditions: Temperature – facility ambient within the range of 16EC to 32EC; Relative humidity – facility ambient up to 95 percent; Atmospheric pressure - facility ambient.

4.5 Sample size. Inspection sample size shall be determined following the guidance contained in ANSI/ASQ Z1.4, at an initial inspection level II (normal). The physical values specified shall apply to the average of determinations made on a sample unit for test purposes as required by the applicable test methods.

4.6 Examinations and tests.

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4.6.1 Composition. The composition of the sleeving shall be identified and certified by the supplier.

4.6.2 Construction. The construction of the sleeving shall be determined visually.

4.6.3 Sleeving dimensions. The sleeving wall thickness shall be measured with a paper micrometer or equivalent instrument. Other sleeving dimensions shall be determined by visual examination and appropriate measurements.

4.6.4 Color. The sleeving shall be visually examined for color.

4.6.5 Braid coverage. Sleeving specimens, each not less than 6 inches in length, shall be carefully cut and fitted over a smooth-surfaced mandrel which has hemispherical ends. The mandrel diameter shall correspond in size to the inside diameter of the sleeving within " 1/64 inch. The sleeving shall be fitted on the mandrel in a manner where it is not stretched (tightly fitted), loosely fitted, compressed, or gathered. The braid coverage shall then be determined by using the following formula:

$$K = 100 \times (2F \mathbf{B} F^2), \quad \text{where:}$$

K = percent of braid coverage
F = N x P x 0.0234 / sin(a)
N = number of ends per carrier
P = picks per inch of sleeving length
a = angle of braid with longitudinal axis of sleeving

4.6.6 Tensile strength. At least six unknotted and jointless cords, each at least 18 inches long, shall be carefully removed from finished sleeving and tested in a suitable machine for tensile breaking strength. The cords shall be stretched in the machine at a rate of 12 " **2** inches per minute until they break. The breaking strength shall be recorded in all cases. Tests in which the specimen breaks at the jaw of the test machine shall be disregarded and those tests shall be repeated with new specimens. The minimum tensile breaking strength shall not be less than the value specified in 2.5. The average breaking strength for the jointless cords shall be calculated using all, but not less than six, successfully tested specimens. Based on similar tests, the supplier shall certify that the minimum breaking strength of fused cord joints conforms to the requirements specified in 2.5.

4.6.7 Fungus resistance. A sleeving specimen shall be tested in a suitable test chamber to assess the extent to which the sleeving will support fungus growth. The chamber shall operate at ambient atmospheric pressure, shall be prevent condensation from dripping on the test item, and shall maintain the temperature and relative humidity in the test chamber at 30EC " 1EC and 95% " 5% for the duration of the test. The duration of the test shall be at least 28 days. Test chamber temperatures and relative

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humidity shall be recorded continuously during the test. Immediately after the completion of the test, the test specimen shall be visually inspected to verify that it did not support fungus growth.

4.6.8 Workmanship. The sleeving shall be visually examined to ensure the sleeving is free of visible defects, dirt, grime, grease, and abraded or damaged surfaces.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Part or identifying number (PIN). The following PIN may be used to identify specific sizes of this sleeving:

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Basic CID number
Dash number (Size - see table II)

TABLE II. Sleeving sizes.

Dash Number	Size	Dash Number	Size
1	1/8	8	3/4
2	3/16	9	7/8
3	1/4	10	1
4	5/16	11	1-1/4
5	3/8	12	1-1/2
6	1/2	13	1-3/4
7	5/8	14	2

Custodian:
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